



Fig. 9: Bluetooth module

a barcode or magnetic strip must be scanned to get information, the RFID device must be scanned to retrieve identifying information.

Bluetooth. It is a wireless technology standard for exchanging data over limited short distances using short-wavelength UHF radio waves in the ISM band (free band for use



Fig. 10: ZigBee (Xbee) module

in communication) from 2.4GHz to 2.485GHz.

This technology, invented by telecom vendor company Ericsson in 1994, was initially pitched as a wireless alternative to RS232 serial data communication.

Bluetooth (Fig. 9) is managed by Bluetooth Special Interest Group (SIG), which has more than 20,000 member companies in areas of telecommunication, computing, networking and consumer electronics. It was standardised as IEEE 802.15.1

but the standard is not maintaining.

There are two classes of Bluetooth devices; class I has a range of up to 100m and class II a range of up to 10m. In applications like general-purpose wireless communication application, class II is mostly used.

ZigBee (XBee). This is used in high-level communication protocols to create personal area networks built from small, low-power digital radios. ZigBee (Fig. 10) is based on IEEE 802.15.4 standard. Its low power consumption limits transmission distances from 10m to 100m line-of-sight, depending on power output and environmental characteristics.

ZigBee devices can transmit data over long distances by passing data through a mesh network of intermediate devices to reach more distant places. These are typically used in low-data-rate applications that require a long battery life. This type of system is commonly used in home or building automation where more than one node is communicating. ●

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